

# JUMANA

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## ACADEMIC CURRICULUM VITAE

### PERSONAL PROFILE

Computer Science and Engineering student at Varendra University with research interests in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Healthcare AI, Agricultural AI, and Internet of Things (IoT). Author and co-author of multiple conference papers focusing on intelligent disease detection, medical image analysis, autism spectrum disorder detection, and smart healthcare systems. Passionate about conducting impactful research and developing AI-driven solutions for real-world challenges.

### RESEARCH INTERESTS

- Machine Learning
- Deep Learning
- Computer Vision
- Artificial Intelligence
- Healthcare AI
- Agricultural AI
- Medical Image Analysis
- Neural Networks
- Internet of Things (IoT)
- Intelligent Systems

### EDUCATION

#### Bachelor of Science (B.Sc.) in Computer Science and Engineering

Varendra University, Bangladesh

- Current Semester: 6th
- Expected Graduation: 2027

### RESEARCH PUBLICATIONS

#### 1. A Custom Neural Network Approach for Leaf Disease Detection Using CNN Features

Conference Paper (IEEE)

**Research Area:** Agricultural AI, Deep Learning, Computer Vision

**Summary:** Developed a custom neural network model utilizing CNN-extracted features to identify plant leaf diseases accurately and support intelligent agricultural monitoring.

## **2. Multiclass Skin Lesion Classification Using EfficientNetB4 and a Custom Feed-Forward Neural Network**

### **Conference Paper (IEEE)**

**Research Area:** Healthcare AI, Medical Image Analysis

**Summary:** Proposed a multiclass skin lesion classification framework using EfficientNetB4 combined with a custom feed-forward neural network for improved diagnostic performance.

## **3. Autism Spectrum Disorder Detection Using a Fine-Tuned VGG16 Model**

### **Conference Paper**

**Research Area:** Healthcare AI, Deep Learning

**Summary:** Investigated autism spectrum disorder detection using transfer learning techniques and a fine-tuned VGG16 architecture.

## **4. Touchless IoT-Based Health Monitoring System for Pregnant Women and Elderly Patients with Real-Time Cloud Integration**

### **Conference Paper**

**Research Area:** IoT, Smart Healthcare Systems

**Summary:** Designed an IoT-enabled healthcare monitoring system for remote patient observation with cloud-based real-time data integration.

## **RESEARCH PROJECTS**

### **Plant Disease Detection System**

- Developed a plant disease detection framework using deep learning techniques.
- Utilized CNN architectures and image preprocessing methods.
- Applied computer vision techniques for disease identification from leaf images.

#### **Technologies:**

Python, TensorFlow, Keras, OpenCV, Deep Learning

### **Skin Lesion Classification System**

- Developed a medical image classification system for skin lesion detection.
- Applied EfficientNetB4 and custom neural network architectures.
- Evaluated model performance using image classification metrics.

#### **Technologies:**

Python, TensorFlow, EfficientNetB4, Deep Learning

### **Autism Spectrum Disorder Detection**

- Developed a deep learning-based autism detection model.
- Applied transfer learning using VGG16 architecture.

- Conducted model training and evaluation on autism-related datasets.

**Technologies:**

Python, TensorFlow, VGG16, Deep Learning

**IoT-Based Healthcare Monitoring System**

- Designed a touchless healthcare monitoring platform.
- Integrated cloud-based monitoring and real-time health data collection.
- Focused on healthcare support for pregnant women and elderly patients.

**Technologies:**

IoT, Cloud Computing, Sensors

**SOFTWARE & AI PROJECTS****CS50AI PageRank**

Implemented Google's PageRank algorithm using graph-based ranking techniques.

**CS50AI TicTacToe**

Developed an AI-powered TicTacToe game using the Minimax algorithm.

**CS50 Minesweeper**

Created a knowledge-based AI agent capable of solving Minesweeper puzzles through logical inference.

**AI50 X Degrees**

Implemented graph search algorithms to determine relationships between entities.

**Flutter Mobile Applications**

Developed mobile applications using Flutter and Dart.

**TECHNICAL SKILLS****Programming Languages**

- Python
- C
- C++
- Java

**Web Technologies**

- HTML
- CSS
- JavaScript

## **Database**

- SQL (Basic)

## **Artificial Intelligence & Machine Learning**

- Machine Learning
- Deep Learning
- Computer Vision
- Convolutional Neural Networks (CNN)
- Transfer Learning
- Neural Networks
- TensorFlow
- Keras
- Image Classification

## **Tools & Platforms**

- Git
- GitHub
- VS Code
- Jupyter Notebook
- Google Colab

## **CONFERENCE PARTICIPATION & CERTIFICATIONS**

### **IEEE Conference Participation Certificates (2)**

Participated in IEEE-affiliated academic conferences and presented research activities.

### **UCICS Conference**

Participant in the UCICS Conference organized by the Department of Computer Science and Engineering, Varendra University.

## **LEADERSHIP & VOLUNTEER EXPERIENCE**

### **Conference Volunteer**

- Served as a volunteer in two academic/research conferences.
- Assisted in event coordination and participant support.
- Collaborated with organizing committees to ensure successful event execution.

### **Debate Activities**

- Active participant in debate and public speaking activities.
- Developed communication, critical thinking, and presentation skills.

## **ACHIEVEMENTS**

- Author/Co-Author of Four Conference Research Papers
- IEEE Conference Participant
- Active Researcher in Artificial Intelligence and Machine Learning
- Developed Multiple AI-Based Research Projects
- Experience in Deep Learning and Computer Vision Applications
- Strong Academic Interest in Research and Innovation

## **LANGUAGES**

- Bengali (Native)
- English (Professional Working Proficiency)